

1D CNN — Architecture Variant Comparison

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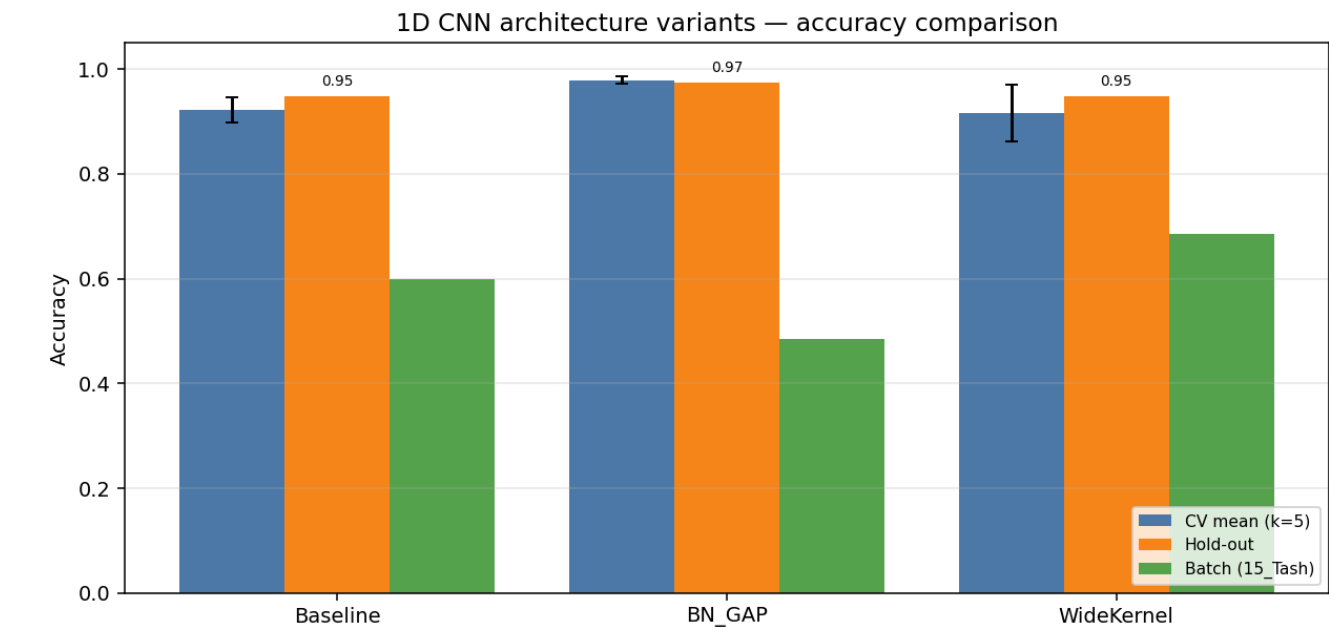
Data & configuration

Training path 1	/home/jestin/ThesisRepo/ML/NewTestData/1_Alan/Dynamic
Training path 2	/home/jestin/ThesisRepo/ML/NewTestData/2_Alex/Dynamic
Training path 3	/home/jestin/ThesisRepo/ML/NewTestData/3_Anghad/Dynamic
Training path 4	/home/jestin/ThesisRepo/ML/NewTestData/4_Daniel/Dynamic
Training path 5	/home/jestin/ThesisRepo/ML/NewTestData/6_Harry/Dynamic
Training path 6	/home/jestin/ThesisRepo/ML/NewTestData/8_Jestin/Dynamic
Training path 7	/home/jestin/ThesisRepo/ML/NewTestData/11_Mansh/Dynamic
Training path 8	/home/jestin/ThesisRepo/ML/NewTestData/12_Marcus/Dynamic
Training path 9	/home/jestin/ThesisRepo/ML/NewTestData/14_StephenV2/Dynamic
Training path 10	/home/jestin/ThesisRepo/ML/NewTestData/17_Raquel/Dynamic
Training path 11	/home/jestin/ThesisRepo/ML/NewTestData/18_Bridgette/Dynamic
Batch-test root	/home/jestin/ThesisRepo/ML/NewTestData/15_Tash/Dynamic
Total trials	376 (train pool 338, hold-out 38)
Classes	7: Double_Lower, Double_Nothing, Double_Pistol_Recoil, Double_Raise, Double_Wiggle, Double_cmere, Drum_Roll
Sensor channels	176 (sequence length 90)
Resample / filter	90 steps · Butterworth cutoff 15.0 Hz, order 4
Normalisation	minmax
CV folds · shuffle	5 · True
Augmentation	on · copies=5 · amplitude_scale, baseline_offset
Epochs · batch size	15 · 16

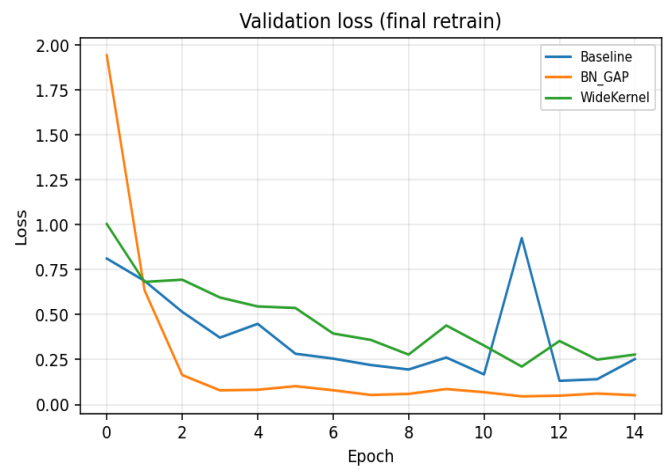
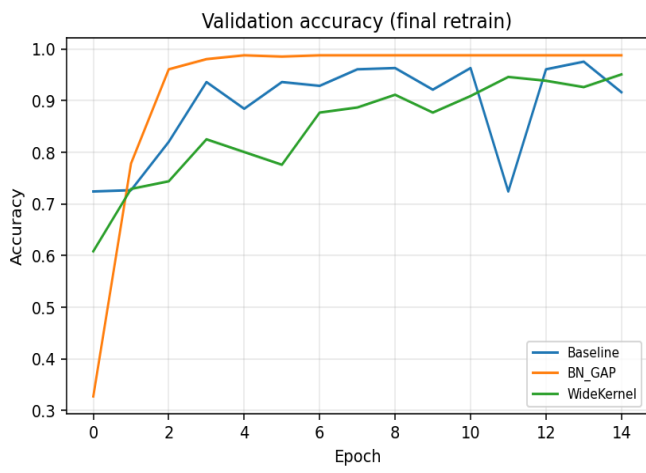
Variant comparison (summary)

Variant	Params	CV mean (k=5)	CV std	Hold-out acc	Hold-out loss	Batch acc	Batch n
Baseline	113,255	0.9230	0.0240	0.9474	0.3895	0.6000	21/35
BN_GAP	57,319	0.9793	0.0073	0.9737	0.2786	0.4857	17/35
WideKernel	131,687	0.9169	0.0540	0.9474	0.1685	0.6857	24/35

Best on hold-out: **BN_GAP** · Best on CV: **BN_GAP**



Training curves (final retrain on full training pool)



Variant: Baseline

Conv1D(32,k=4)→Pool→Conv1D(64,k=4)→Pool→Flatten→Dense(64)→Dropout(0.3)→Softmax

Trainable params	113,255
CV mean ± std	0.9230 ± 0.0240
Hold-out test acc / loss	0.9474 · 0.3895
Batch-test acc	0.6000 (21/35)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1620	68	0.9265	0.9853	0.9265
2	1620	68	0.9265	0.9853	0.9265
3	1620	68	0.9265	0.9265	0.9265
4	1626	67	0.8806	0.9104	0.8806
5	1626	67	0.9552	0.9701	0.9552

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	5
Double_Nothing	1.000	0.600	0.750	5
Double_Pistol_Recoil	1.000	1.000	1.000	6
Double_Raise	0.857	1.000	0.923	6
Double_Wiggle	0.833	1.000	0.909	5
Double_cmere	1.000	1.000	1.000	6
Drum_Roll	1.000	1.000	1.000	5
macro avg	0.956	0.943	0.940	38
weighted avg	0.956	0.947	0.943	38

Batch test per class (15_Tash)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Nothing	0	5	0.000
Double_Pistol_Recoil	5	5	1.000
Double_Raise	5	5	1.000
Double_Wiggle	4	5	0.800
Double_cmere	2	5	0.400
Drum_Roll	0	5	0.000
Overall	21	35	0.600

Variant: BN_GAP

Conv+BN+ReLU ×3 with GlobalAveragePooling1D head — fewer params, less overfit

Trainable params	57,319
CV mean ± std	0.9793 ± 0.0073
Hold-out test acc / loss	0.9737 · 0.2786
Batch-test acc	0.4857 (17/35)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1620	68	0.9853	0.9853	0.9853
2	1620	68	0.9853	0.9853	0.9853
3	1620	68	0.9706	1.0000	0.9706
4	1626	67	0.9701	0.9701	0.9701
5	1626	67	0.9851	0.9851	0.9851

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	5
Double_Nothing	1.000	0.800	0.889	5
Double_Pistol_Recoil	1.000	1.000	1.000	6
Double_Raise	1.000	1.000	1.000	6
Double_Wiggle	1.000	1.000	1.000	5
Double_cmere	0.857	1.000	0.923	6
Drum_Roll	1.000	1.000	1.000	5
macro avg	0.980	0.971	0.973	38
weighted avg	0.977	0.974	0.973	38

Batch test per class (15_Tash)

Class	Correct	Total	Accuracy
Double_Lower	0	5	0.000
Double_Nothing	1	5	0.200
Double_Pistol_Recoil	3	5	0.600
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	2	5	0.400
Drum_Roll	1	5	0.200
Overall	17	35	0.486

Variant: WideKernel

Conv1D(32,k=8)→Pool→Conv1D(64,k=8)→Pool→Flatten→Dense(64)→Dropout(0.3)→Softmax

Trainable params	131,687
CV mean ± std	0.9169 ± 0.0540
Hold-out test acc / loss	0.9474 · 0.1685
Batch-test acc	0.6857 (24/35)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1620	68	0.9412	0.9412	0.9412
2	1620	68	0.9853	0.9853	0.9853
3	1620	68	0.9118	0.9265	0.9118
4	1626	67	0.9254	0.9254	0.9254
5	1626	67	0.8209	0.9851	0.8209

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	5
Double_Nothing	1.000	0.600	0.750	5
Double_Pistol_Recoil	1.000	1.000	1.000	6
Double_Raise	0.857	1.000	0.923	6
Double_Wiggle	0.833	1.000	0.909	5
Double_cmere	1.000	1.000	1.000	6
Drum_Roll	1.000	1.000	1.000	5
macro avg	0.956	0.943	0.940	38
weighted avg	0.956	0.947	0.943	38

Batch test per class (15_Tash)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Nothing	1	5	0.200
Double_Pistol_Recoil	5	5	1.000
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	2	5	0.400
Drum_Roll	1	5	0.200
Overall	24	35	0.686